Tissue Damage Analysis

8-hour simulation | STII model (Linder-Ganz & Gefen 2007)

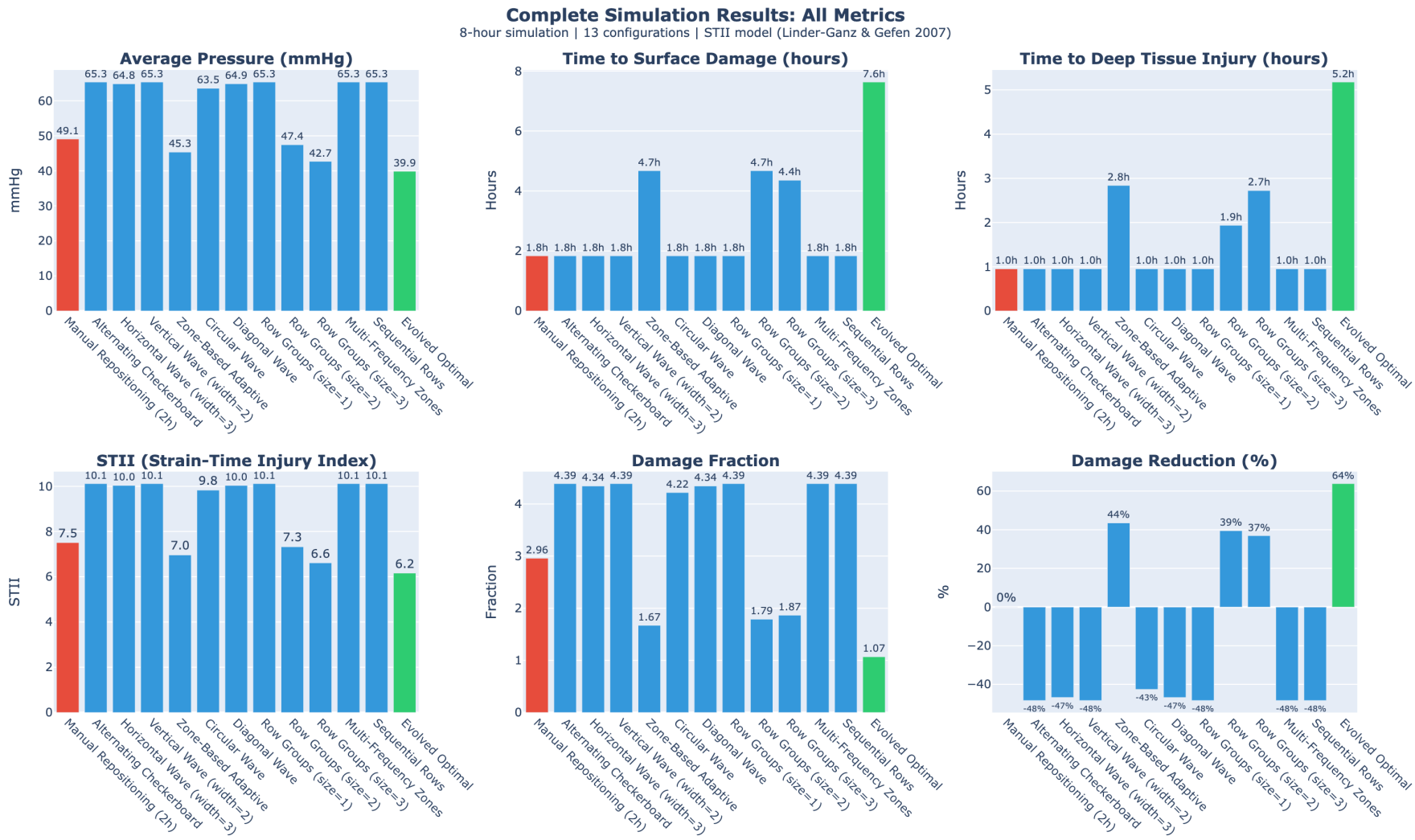

Table 1: Raw Simulation Data

Configuration	Pressure	Surface (h)	DTI (h)	STII	Damage
Manual Repositioning (2h)	49.07	1.8	1.0	7.51	2.96
Alternating Checkerboard	65.34	1.8	1.0	10.12	4.39
Horizontal Wave (width=3)	64.84	1.8	1.0	10.04	4.34
Vertical Wave (width=2)	65.34	1.8	1.0	10.12	4.39
Zone-Based Adaptive	45.34	4.7	2.8	6.96	1.67
Circular Wave	63.55	1.8	1.0	9.83	4.22
Diagonal Wave	64.86	1.8	1.0	10.03	4.34
Row Groups (size=1)	65.34	1.8	1.0	10.12	4.39
Row Groups (size=2)	47.44	4.7	1.9	7.33	1.79
Row Groups (size=3)	42.69	4.4	2.7	6.61	1.87
Multi-Frequency Zones	65.34	1.8	1.0	10.12	4.39
Sequential Rows	65.34	1.8	1.0	10.12	4.39
Evolved Optimal	39.87	7.6	5.2	6.16	1.07

Table 2: Percent Change from Baseline

Configuration	Pressure %	DTI %	STII %	Damage Red %
Manual Repositioning (2h)	Baseline	Baseline	Baseline	0.0
Alternating Checkerboard	+33.1	+0.0	+34.7	-48.4
Horizontal Wave (width=3)	+32.1	+0.0	+33.6	-46.8
Vertical Wave (width=2)	+33.1	+0.0	+34.7	-48.4
Zone-Based Adaptive	-7.6	+198.2	-7.3	43.5
Circular Wave	+29.5	+0.0	+30.9	-42.6
Diagonal Wave	+32.2	+0.0	+33.6	-46.8
Row Groups (size=1)	+33.1	+0.0	+34.7	-48.4
Row Groups (size=2)	-3.3	+103.5	-2.4	39.5
Row Groups (size=3)	-13.0	+186.0	-11.9	36.9
Multi-Frequency Zones	+33.1	+0.0	+34.7	-48.4

Sequential Rows	+33.1	+0.0	+34.7	-48.4
Evolved Optimal	-18.7	+443.9	-17.9	63.8

Best Configuration: Evolved Optimal

Metric	Value
Pressure Reduction	-18.7%
DTI Improvement	+443.9%
STII Reduction	-17.9%
Damage Reduction	63.8%
DTI Efficiency	544%